

HIGHLIGHTS for Review

The problem addressed by this paper is the routing of care worker teams that is similar to the real-world multiple traveling salesman problem. This

- a collection of patients need to be visited by health care workers.
- Some objectives are to reduce the total cost and traveling time when visiting geographically distributed patients with satisfaction of care workers as well as patients.

The problem solving approach proposed is a soft computing approach which considers:

- an attempt to use a hybridization of two soft computing known techniques: quantum computing and genetic algorithm in home health care scheduling.
- quantum computing extends computer capacity to more than what was traditionally and for long time known as strict frontiers of storage capacity and process computing ability.
- an optimization tool derived from this hybridization is a quantum inspired genetic algorithm which has the advantage to dig sensitively around a reached minima to explore other possible optimization solutions without falling in as local minima.

The implementation of this soft computing proposed approach used :

- Matlab R2017b release as programming language.
- A laptop of 4 G RAM and 700 G disk storage capacity with Intel ® Core (TM @ i5-3210M CPU 2.5 GHZ) under windows 7 Home 64 bits.
- A comparative study between three algorithms using fictitious data (traditional genetic algorithm and quantum inspired genetic algorithm) was carried out in the framework of this paper and showed superiority of quantum evolutionary approach in solving home health care routing problem.
- The future improvement will consider real data and some ignored parameters like time windows and other for the simplicity of application.